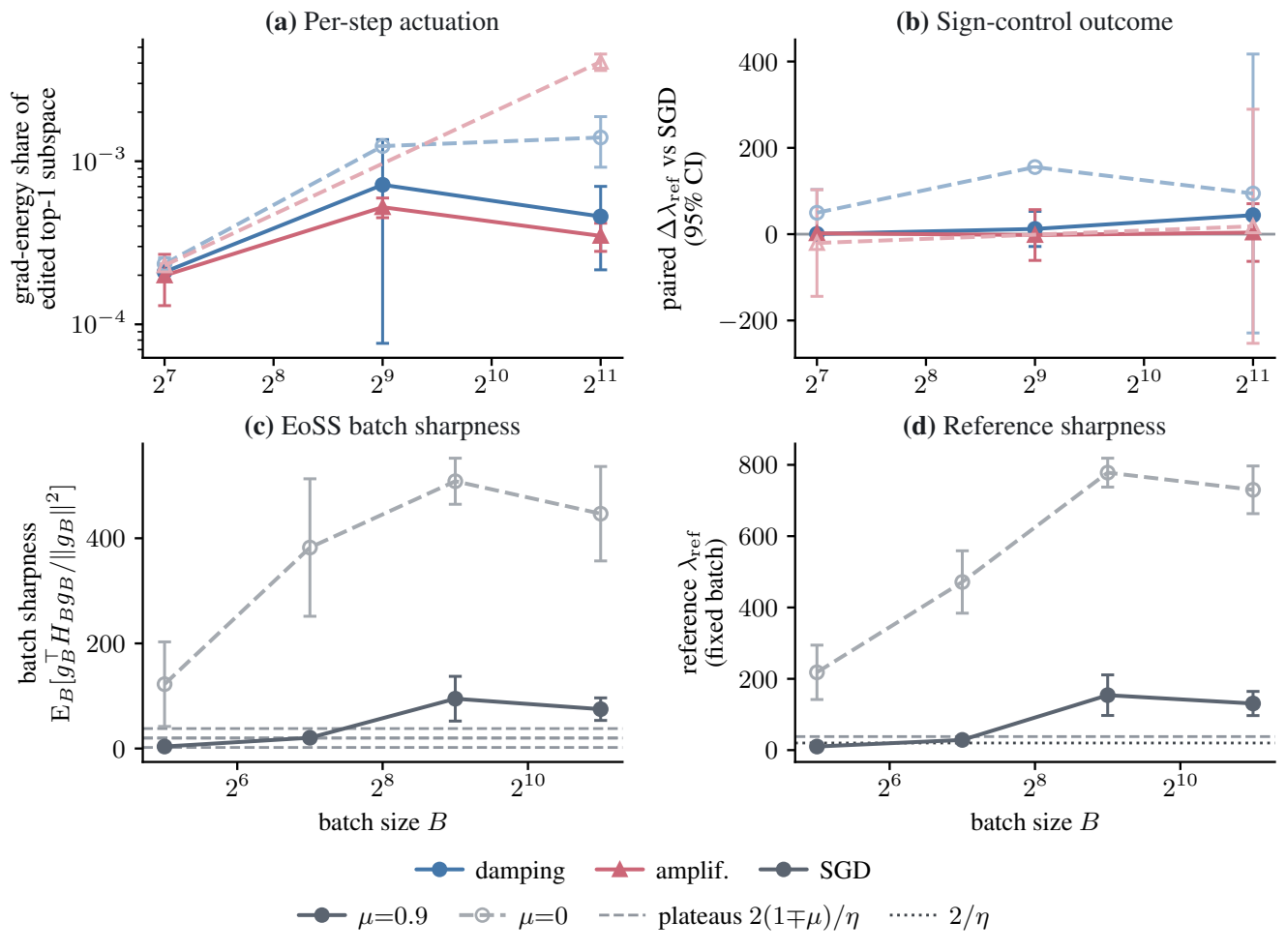

(a) Per-step actuation
grad-energy share of edited top-1 subspace
10^{-3}
10^{-4}
2^7
2^8
2^9
2^{10}
2^{11}
(b) Sign-control outcome
paired $\Delta\lambda_{\mathrm{ref}}$ vs SGD (95% CI)
400
200
0
−200
(c) EoSS batch sharpness
batch sharpness $\mathrm{E}_B[g_B^\top H_B g_B / \|g_B\|^2]$
400
200
0
2^6
batch size B
(d) Reference sharpness
reference λ_{ref} (fixed batch)
800
600
400
200
0
damping
amplif.
SGD
μ=0.9
μ=0
plateaus $2(1\mp\mu)/\eta$
$2/\eta$